

MATH 5023

Project 8: Multivariable Taylor Theorem

Main source. C. H. Edwards, Jr., *Advanced Calculus of Several Variables*, Chapter II.

Notation from Section 2

- Let $U \subset \mathbb{R}^n$ be open and let $f : U \rightarrow \mathbb{R}$.
- $f \in C^k(U)$ means all iterated partial derivatives of order at most k exist and are continuous on U .
- $D_i f = \frac{\partial f}{\partial x_i}$. Thus $D_{i_1} D_{i_2} \cdots D_{i_q} f$ means repeated partial differentiation.
- For a fixed direction $h = (h_1, \dots, h_n)$, the directional derivative operator is

$$D_h = h_1 D_1 + \cdots + h_n D_n.$$

- $D_h^r f(a)$ means apply the same directional derivative r times and then evaluate at a .

Examples for the Notation

If $f = f(x, y, z)$, then $D_1f = f_x$, $D_2f = f_y$, and $D_3f = f_z$. For instance, D_1D_2f means taking two partial derivatives, one with respect to y and one with respect to x .

If $f(x, y) = x^2 + y^2$, $h = (h_1, h_2)$, and $a = (0, 0)$, then

$$D_h = h_1D_1 + h_2D_2, \quad D_h^2f(a) = 2h_1^2 + 2h_2^2.$$



Multivariable Taylor Theorem

Let $k \geq 0$, $a \in \mathbb{R}^n$, $h \in \mathbb{R}^n$, and

$$L = \{a + th : 0 \leq t \leq 1\}.$$

Suppose $U \subset \mathbb{R}^n$ is open, $L \subset U$, and $f : U \rightarrow \mathbb{R}$ has continuous partial derivatives up to order $k + 1$. Then there exists $\xi \in L$ such that

$$f(a + h) = \sum_{r=0}^k \frac{D_h^r f(a)}{r!} + \frac{D_h^{k+1} f(\xi)}{(k + 1)!}.$$



What the Formula Is Saying

The Taylor polynomial is

$$P_k(h) = \sum_{r=0}^k \frac{D_h^r f(a)}{r!},$$

and the remainder is

$$R_k(h) = f(a + h) - P_k(h) = \frac{D_h^{k+1} f(\xi)}{(k + 1)!}.$$

The point ξ is somewhere on the line segment from a to $a + h$. Thus the formula says: the value of f at $a + h$ equals a polynomial built from derivative information at a , plus one exact error term.



Proof Idea: Reduce to One Variable

Define

$$g : [0, 1] \rightarrow \mathbb{R}, \quad g(t) = f(a + th).$$

Then $g(0) = f(a)$ and $g(1) = f(a + h)$. By the chain rule,

$$g'(t) = D_1 f(a + th)h_1 + \cdots + D_n f(a + th)h_n = D_h f(a + th).$$

Repeating the same idea gives

$$g^{(r)}(t) = D_h^r f(a + th).$$



Proof Idea: Substitute into Taylor’s Formula

Apply the one-variable Taylor theorem to g from 0 to 1. There is some $c \in (0, 1)$ such that

$$g(1) = \sum_{r=0}^k \frac{g^{(r)}(0)}{r!} + \frac{g^{(k+1)}(c)}{(k+1)!}.$$

Since $g^{(r)}(0) = D_h^r f(a)$ and $g^{(k+1)}(c) = D_h^{k+1} f(a + ch)$, set $\xi = a + ch$. Then $\xi \in L$, and the multivariable formula follows.



Example: Second-Order Expansion

Let $f(x,y) = x^2 + xy + y^3$, expand at $a = (0,0)$, and write $h = (u,v)$. Since

$$f(0,0) = 0, \qquad f_x(0,0) = 0, \qquad f_y(0,0) = 0,$$

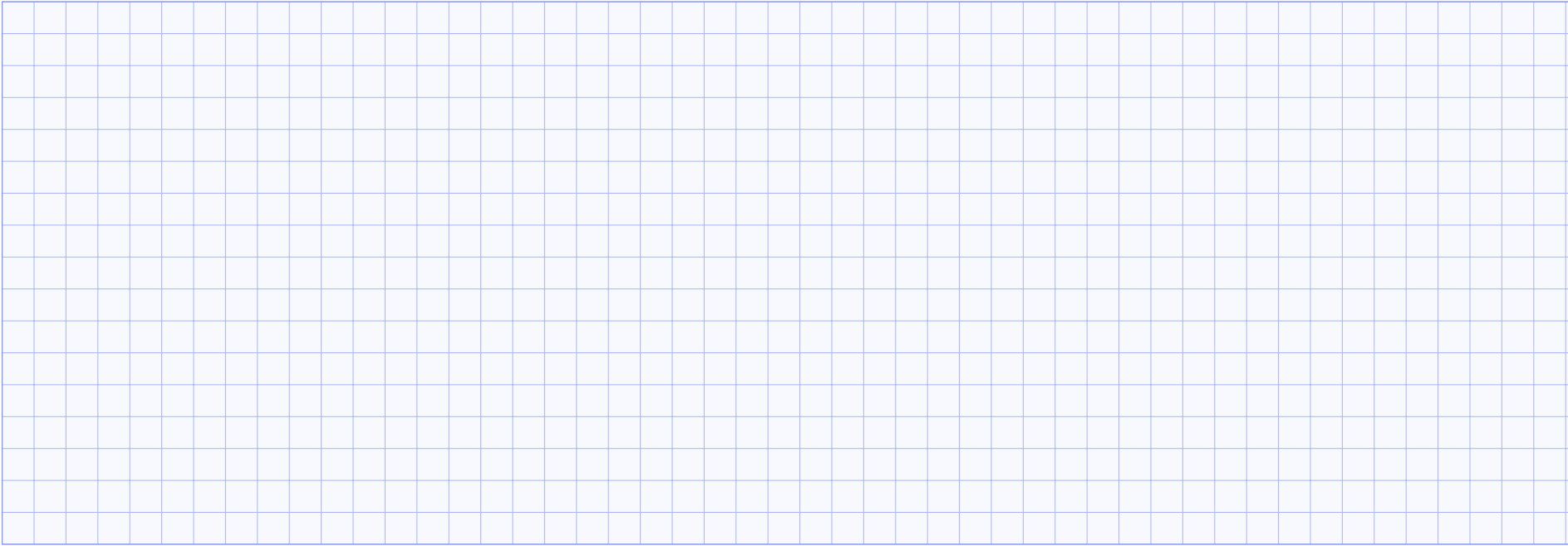
and

$$f_{xx}(0,0) = 2, \qquad f_{xy}(0,0) = 1, \qquad f_{yy}(0,0) = 0,$$

the second-degree Taylor polynomial is

$$P_2(u,v) = u^2 + uv.$$

But $f(u,v) = u^2 + uv + v^3$, so the remainder is $R_2(u,v) = v^3$.



Application: Error Estimate

For $f(x, y) = e^{x+y}$, $a = (0, 0)$, and $h = (x, y)$, we have

$$P_2(h) = 1 + (x + y) + \frac{(x + y)^2}{2}.$$

Taylor's theorem says that for some $\xi = (\xi_1, \xi_2)$ on the line segment from $(0, 0)$ to (x, y) ,

$$e^{x+y} = 1 + (x + y) + \frac{(x + y)^2}{2} + \frac{e^{\xi_1 + \xi_2}(x + y)^3}{6}.$$

If $|x| \leq 0.1$ and $|y| \leq 0.1$, then $|\xi_1 + \xi_2| \leq 0.2$, so the error is bounded by

$$|R_2(h)| \leq \frac{e^{0.2}|x + y|^3}{6}.$$



Application: Critical Points

Taylor's theorem can also classify local behavior near a critical point. For example,

$$f(x, y) = x^2 + y^2 + \cos x.$$

At $(0, 0)$, $f_x(0, 0) = 0$ and $f_y(0, 0) = 0$. Using

$$\cos x = 1 - \frac{x^2}{2} + R_2(x),$$

we get

$$f(x, y) = 1 + \frac{x^2}{2} + y^2 + R_2(x).$$

The quadratic part $\frac{x^2}{2} + y^2$ is positive except at $(0, 0)$, so the function has a local minimum there.



Connection with Newton's Method

Newton's method uses the quadratic Taylor approximation

$$f(a+h) \approx f(a) + \nabla f(a) \cdot h + \frac{1}{2} h^T H(a) h.$$

For

$$f(x, y) = (x-1)^2 + (y+2)^2,$$

we have $\nabla f(0, 0) = (-2, 4)$ and $H = 2I$. Starting at $x_0 = (0, 0)$,

$$x_1 = x_0 - H^{-1} \nabla f(x_0) = (0, 0) - \frac{1}{2}(-2, 4) = (1, -2).$$

This reaches the minimum in one step because the function is already quadratic.

